Kannan S

Software Engineer

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Profile Summary

Software Engineer building web products with JavaScript, TypeScript, and modern frontend and backend technologies. Builds maintainable features across user interfaces, APIs, authentication, payments, AI integrations, real-time systems, and data workflows, with a focus on security and real-world requirements. Works across the product stack from interface design and backend development to deployment and release support.

Technical Skills

Languages: JavaScript, TypeScript, C, HTML, CSS
Libraries & Frameworks: Node.js, Express.js, React.js, Tailwind CSS, Bootstrap, React Router, EJS
Databases & Caching: MongoDB, Mongoose, MySQL, Redis, Firebase Firestore
AI & Integrations: Groq AI (Llama 3.3 70B), Google Gemini API, LLM API Integration, pdf-parse, Multer
Tools & Platforms: Git, GitHub, Postman, Nginx, PM2, AWS EC2, AWS S3, Docker
Design & Prototyping: Figma
Concepts: REST APIs, WebRTC, Socket.io, CI/CD, OOP, SOLID Principles, Data Structures, Algorithms

Professional Experience

Full-Stack Developer Trainee *2025 – Present*

Industry-Oriented Software Development Program

- Completed intensive training in **JavaScript, TypeScript, Node.js, React.js, Express.js, and MongoDB**, with hands-on implementation of full-stack applications.
- Practiced **Data Structures and Algorithms**, including arrays, strings, linked lists, recursion, searching, sorting, and problem-solving techniques, with emphasis on time and space complexity.
- Developed end-to-end projects involving **REST APIs**, database design, authentication, payment integrations, real-time communication, and cloud deployment.
- Participated in weekly technical evaluations and project reviews conducted by experienced industry professionals, incorporating feedback on code quality, problem-solving, architecture, and software engineering practices.

Apprentice Trainee *2024*

Bharat Sanchar Nigam Limited (BSNL)

- Gained practical exposure to organizational operations, customer engagement, and business service workflows within the telecommunications sector.
- Assisted with customer interactions, service-related inquiries, and communication of telecom products and service offerings.
- Supported marketing activities, promotional initiatives, and coordination of customer outreach operations.
- Developed professional skills in communication, client handling, teamwork, and understanding customer requirements in a business environment.

Projects

zenCode: Real-Time Coding Interview Platform Live | GitHub

React, Node.js, Express.js, MongoDB, Redis, Docker, Socket.io, WebRTC, TypeScript, AWS

- Engineered a **real-time collaborative workspace** combining **WebRTC** peer-to-peer video/audio, a **Socket.io-synchronized Monaco Editor** shared between mentor and candidate, **in-session chat** with typing indicators, and live code execution result broadcasting — all within a single resizable panel layout.
- Built an isolated **Docker-based code execution engine** (Python/Node.js) with Base64 command tunneling for shell injection prevention, **Redis rate-limiting** (20 req/min), and a marker-based **hidden test case evaluator** achieving sub-second execution times.
- Architected a secure **Express.js & MongoDB** backend following **layered clean architecture** (Routes → Controllers →

Services → Repositories), with **JWT + Google OAuth** authentication and **Redis-backed** session and OTP management.

- Integrated the **Google Gemini API** to deliver context-aware, progressive **AI programming hints** per problem without exposing solutions, and built a **Stripe subscription billing system** with **webhook-based lifecycle management** to gate premium content access.
- Configured an **automated CI/CD pipeline** via **GitHub Actions** to build and deploy the full-stack application to **AWS EC2**, with **Nginx** as a reverse proxy, **PM2** for process management, and **AWS S3** for mentor avatar storage via presigned URLs.

SkillMount: AI-Powered Resume Analysis & Career Intelligence Platform GitHub

React, Node.js, Express.js, MongoDB, TypeScript, Groq AI

- Built a **modular monolith** backend with 7 independent feature modules designed for microservices migration.
- Integrated **Groq AI (Llama 3.3 70B)** to analyze resumes and return structured JSON with ATS score, strengths, weaknesses, and improvement suggestions.
- Implemented **skill gap detection** by diffing resume skills against target job role datasets, with mapped course recommendations.
- Built **career pivot prediction** engine scoring all job roles by skill match percentage with transition difficulty labels.
- Developed **CV tailoring** feature using AI to rewrite resume sections per job description, with auto-apply stub for future Chrome extension integration.
- Implemented **JWT-based auth** with role-based access — users manage their own resumes, admins review all resumes and approve career paths via admin vault.

Chapterless: E-Commerce Bookstore Platform Live | GitHub

Node.js, Express.js, MongoDB, EJS, Bootstrap, Razorpay

- Engineered a **full-stack e-commerce platform** with inventory-aware ordering, dynamic pricing, and scalable product catalog architecture.
- Built a **secure authentication system** using session-based auth, OTP verification, and **role-based access control** for admin/user separation.
- Integrated **Razorpay payment gateway** with server-side **signature verification**, ensuring transaction integrity and preventing tampered payments.
- Designed a **complex order and refund system** handling lifecycle states, partial cancellations, returns, and **payment-method-aware wallet refunds**.
- Developed a **promotion and incentive engine** combining coupons, category/product offers, and a **referral system** with wallet-based rewards and usage constraints.

Automatic Self-Parking Car *Sree Narayana Polytechnic College, 2023*

C, Arduino, Ultrasonic Sensor, IR Sensors, Servo Motors, H-Bridge Driver

- Led a team in designing and developing an **autonomous self-parking car** using Arduino, ultrasonic and IR sensors, servo motors, and H-bridge motor drivers.
- Implemented **parking logic in C** for real-time obstacle detection, distance measurement, and motor control with precise sensor-to-actuator signal handling.
- Designed and integrated **hardware components**, ensuring reliable interaction between sensors, microcontroller, and motor systems across all development phases.
- Developed and tested **control sequences** for accurate parking alignment and collision avoidance, delivering a fully functional prototype.

Education

Bachelor of Computer Applications (BCA) 2026 – Present
Sree Narayana Guru Open University (SGOU)

Diploma in Electronics Engineering 2020 – 2023
Sree Narayana Polytechnic College, Kerala